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Fossil Fuel Prices and Capital Cost: A Machine Learning Driven Study on Energy Transition

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An AI workflow connects energy-price signals with financing scenarios.
Regional fit differs, wind and solar fit most closely, and time-split checks show where price signals add information beyond a trend.

01 Question and data

How do fossil fuel prices and inflation relate to energy-project WACC across China, Europe, and the Middle East?

3 regions 9 technologies 120 monthly rows

Historical price and macroeconomic inputs are paired with 2026–2035 WACC projections from Calcaterra et al. (2024). Coal and gas generation are conventional-technology comparisons.

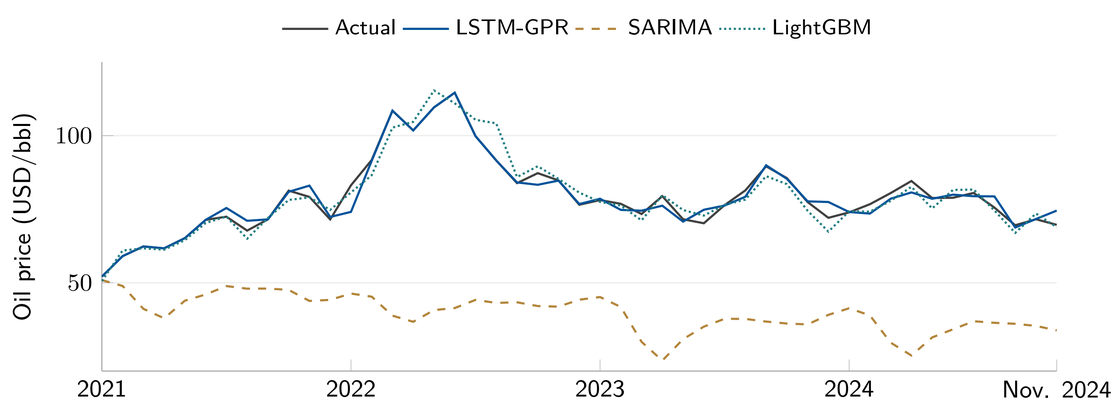
02 Three-stage framework

1 Fuel-price forecasts
CNN, LSTM/GRU, attention, and Gaussian-process smoothing

2 Regional inflation
LSTM + feed-forward network; five-model ensemble

3 WACC comparisons
AutoGluon regressions by region, fuel, and technology

03 Archived price comparison



LSTM-GPR: $R^2 = 0.963$, MAPE 1.86%, RMSE 2.416 USD/bbl. Metrics were recomputed from 47 saved monthly comparison outputs; this is separate from the fixed-origin baseline exercise.

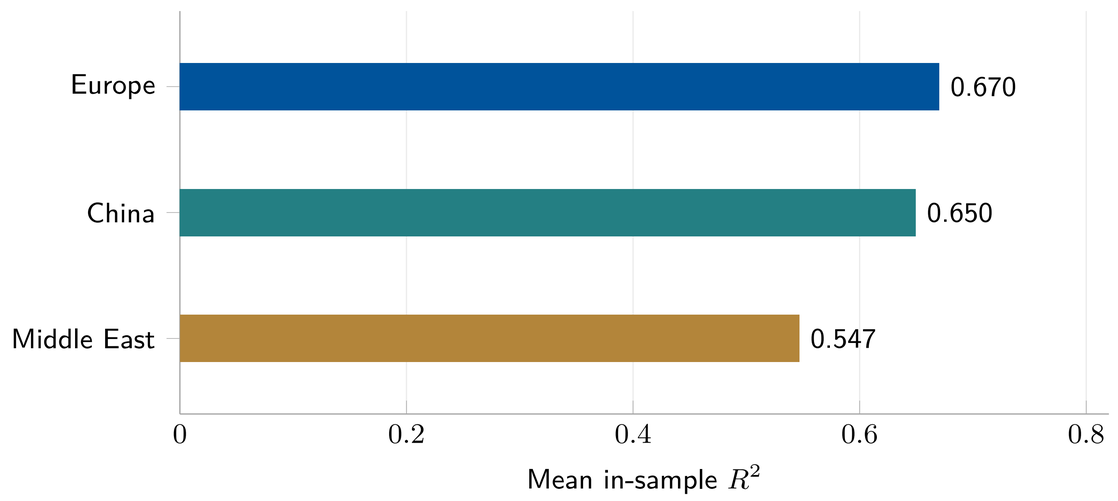
04 Open science and contact

Code + materials

Dataset snapshot

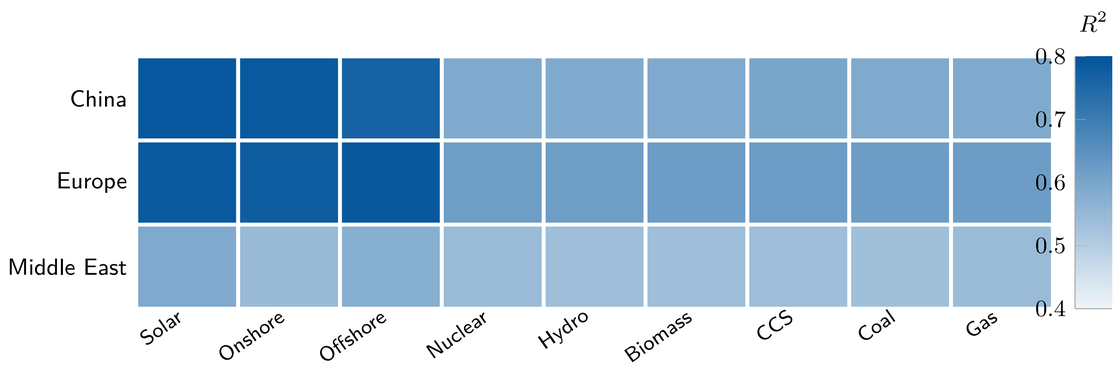
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Editable LaTeX sources and corrected diagnostics are in the GitHub repository.

05 Regional model fit



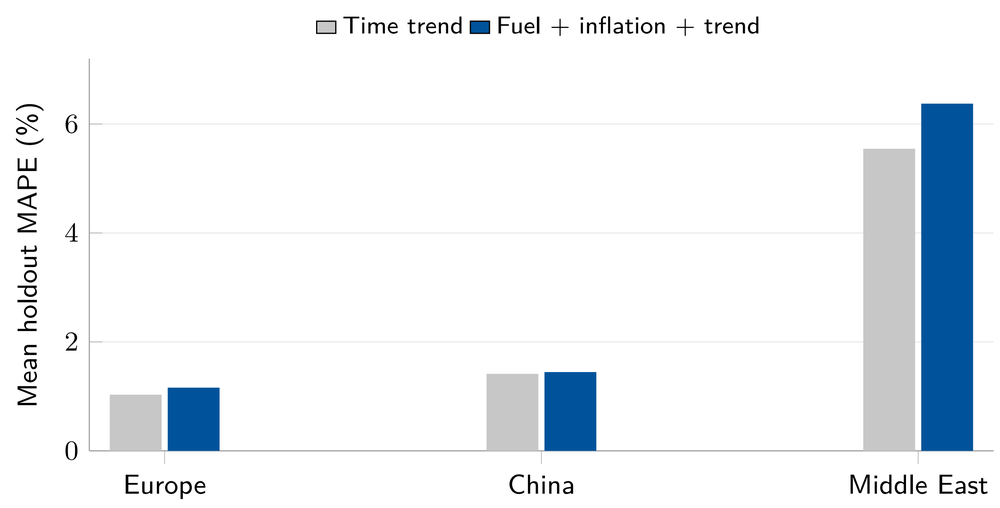
Europe and China have stronger original in-sample fit. These R^2 values describe the joint fuel-price and inflation regressions; they do not measure price-shock magnitude.

06 Technology differences



Mean R^2: solar 0.718; offshore wind 0.706; onshore wind 0.700. Other technologies average around 0.58. Cells average across fuels.

07 Chronological checks



Ridge holdout, 2033–2035. Fuel + inflation + trend improves out-of-sample R^2 over a time trend in China 8/27, Europe 6/27, and Middle East 6/27 cases.

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Selected references

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Calcaterra et al., *Nature Energy* 9, 1241–1251 (2024). doi:10.1038/s41560-024-01606-7.
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OpenAI Codex assisted reference checks, consistency revisions, figure-label corrections, and correction of the sensitivity-analysis code.